

Change Detection in Satellite Imagery Using Siamese Networks:

A Case Study on Pakistani Landscapes

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Computer Vision Course — Final Project

1. Introduction

Urban expansion, deforestation, and climate-driven disasters like flooding are rapidly reshaping landscapes across Pakistan and the broader developing world. Timely, affordable monitoring of these changes remains a critical operational challenge. This project addresses **satellite image change detection** — the task of automatically identifying regions that have undergone meaningful transformation between two time-separated satellite images of the same location.

Manual monitoring by urban planners, disaster response teams, and environmental agencies is slow and costly. An automated computer vision system can flag changes at scale within seconds, enabling near-real-time situational awareness. This is particularly relevant for Pakistan, where rapid urbanisation in cities like Lahore and recurrent flood disasters — most recently the catastrophic 2022 Indus River floods — demand scalable Earth observation pipelines.

We train a **Siamese ResNet-50 network with a Feature Pyramid Network (FPN) decoder** on the public LEVIR-CD building-change dataset, then evaluate qualitatively on real Sentinel-2 imagery over two Pakistani regions. Our system achieves **F1 = 0.906** and **IoU = 0.828** on the LEVIR-CD test set — both substantially exceeding our target thresholds — while surfacing important lessons about domain transfer, metric correctness, and training stability.

2. Related Work

The foundational reference for fully convolutional change detection is **Daudt et al. (2018)**, who introduced FC-EF and FC-Siam-Diff/Conc — the first end-to-end trainable Siamese networks for pixel-wise change detection. Their concatenation-based variant (FC-Siam-Conc) forms the conceptual backbone of our approach.

Chen et al. (2021) proposed BIT (Bitemporal Image Transformer), replacing convolutional encoders with vision transformers to better capture long-range dependencies. Transformer-based methods have since dominated LEVIR-CD leaderboards, though they require substantially more compute and data. We deliberately use a convolutional architecture to demonstrate that strong results are achievable without transformer-scale resources.

The **LEVIR-CD benchmark** (Chen et al., 2020) provides the community standard for building-level change detection, enabling direct comparison of our results with published baselines. WHU-CD (Ji et al., 2019) is a complementary dataset we considered as a fallback.

On the decoder side, **Feature Pyramid Networks** (Lin et al., 2017) introduced the multi-scale skip-connection decoder that we adapt. FPN-style decoders have proven effective for dense prediction tasks where both semantic context (deep features) and spatial precision (shallow features) are required simultaneously.

For the class-imbalance problem inherent in change detection — typically only 5–15% of pixels show change — **Dice Loss** (Milletari et al., 2016) and its combination with Binary Cross-Entropy has become a standard practice in medical and remote sensing segmentation, which we adopt here.

3. Data

3.1 LEVIR-CD (Primary Training Dataset)

The **LEVIR-CD** dataset consists of 637 pairs of high-resolution (1024×1024 pixel) before/after satellite image patches with corresponding binary change masks. Images focus on building construction and demolition in urban areas, with pixel labels: 0 = no change, 255 = change. The dataset exhibits significant **class imbalance** — approximately 10% of pixels show change — which directly influenced our choice of loss function.

Split	Image Pairs	Approx. Tiles (256×256)	Change Pixel %
Train	445	~11,125	~10%
Validation	64	~1,600	~10%
Test	128	~3,200	~10%

3.2 Preprocessing & Tiling

Full 1024×1024 images exceed GPU memory for batch training. We tile each image pair into 256×256 patches using a stride of 192 pixels (64-pixel overlap), yielding 25 overlapping tiles per image pair. The overlap ensures every region of the image is seen from multiple slightly shifted perspectives, improving coverage of objects near tile boundaries.

3.3 Data Augmentation

We apply a two-category augmentation strategy during training only:

- **Spatial augmentations (synchronised):** Horizontal flip, vertical flip, random 90° rotation, ShiftScaleRotate, and GridDistortion — applied identically to both the before (A) and after (B) images to preserve spatial alignment.

- **Colour jitter (independent):** Brightness, contrast, and hue shifts applied independently to A and B. This is a deliberate design choice: real satellite images acquired years apart exhibit different seasonal colours, atmospheric conditions, and sensor calibrations. Applying independent colour jitter teaches the model to ignore these temporal colour shifts and focus on structural, geometric change.

3.4 Real-World Pakistani Imagery (Qualitative Test)

Beyond the LEVIR-CD benchmark, we download free Sentinel-2 true-colour (RGB) imagery at 10 m/pixel resolution from the ESA Copernicus Data Space for two Pakistani regions:

- **Lahore Urban Periphery:** 74.20–74.45°E, 31.35–31.60°N — comparing 2019 vs. 2024 imagery to capture five years of urban expansion.
- **2022 Indus River Floods:** 68.0–70.0°E, 27.0–28.5°N (Sindh / South Punjab) — pre-flood vs. post-flood imagery to identify inundated areas.

These serve as qualitative demonstrations of real-world utility; they are not part of the quantitative evaluation since ground-truth masks are unavailable.

4. Methods

4.1 Siamese Network Architecture

Our model follows the **Siamese paradigm**: two branches sharing identical weights process the before and after images in parallel, producing comparable feature representations that can then be directly compared. The shared weights are critical — they force the network to encode both images using the same learned feature 'language', ensuring that differences in the feature space reflect genuine scene changes rather than encoding artefacts.

4.2 Encoder: ResNet-50

We use a ResNet-50 backbone pre-trained on ImageNet as the shared encoder, loaded via the segmentation-models-pytorch library. The encoder produces a five-level feature pyramid:

Level	Channels	Spatial Size	What It Captures
1	64	64×64	Edges and low-level textures
2	64	64×64	Low-level spatial patterns
3	128	32×32	Object parts and mid-level features
4	256	16×16	Shapes and structural context
5	512	8×8	High-level semantic context

ImageNet pretraining provides a powerful initialisation — the encoder already understands visual structure before seeing a single satellite image, accelerating convergence significantly.

4.3 Feature Comparison: Concatenation

After both images are encoded, features from corresponding levels are combined. We evaluated two strategies: **subtraction** ($F_A - F_B$) and **concatenation** ($[F_A \parallel F_B]$). We use concatenation, as it doubles the channel dimension at each scale and allows the FPN decoder to learn its own comparison function — capturing both magnitude and direction of change at each feature channel. Subtraction, while faster and lighter, destroys the sign of individual channel differences. Our experiments confirm concatenation yields higher F1 (see Section 5).

4.4 Decoder: FPN-Style Reconstruction

The deepest concatenated feature map is only 8×8 pixels; the output must be 256×256 . The FPN decoder achieves this through four progressive upsample-and-merge blocks:

- Bilinear upsample $\times 2$
- Concatenate with the corresponding encoder skip connection at the same spatial scale
- Two Conv–BatchNorm–ReLU layers to process the merged features
- Repeat until 128×128 ; final 1×1 Conv $\rightarrow$ bilinear upsample $\rightarrow 256 \times 256 \rightarrow$ Sigmoid $\rightarrow$ threshold at 0.5

Skip connections are essential: without them, upsampling from 8×8 produces blurry, spatially imprecise outputs. The connections bring back the fine-grained spatial detail from earlier encoder layers.

4.5 Loss Functions

Change detection suffers from severe class imbalance ($\sim 90\%$ no-change pixels). A model that predicts nothing achieves 90% pixel accuracy while being completely useless. We address this with a **combined BCE + Dice loss**:

- **BCE Loss with pos_weight = 8.0**: Standard per-pixel binary cross-entropy. The positive weight of 8.0 means a missed change pixel contributes $8 \times$ more to the loss than a missed no-change pixel, directly penalising the lazy all-zeros prediction.
- **Dice Loss**: Measures region-level overlap: $1 - (2 \times |\text{Pred} \cap \text{GT}|) / (|\text{Pred}| + |\text{GT}|)$. If the model predicts nothing, Dice = 1.0 (maximum penalty). This directly prevents region-level collapse.

Total Loss = $0.5 \times \text{BCE} + 0.5 \times \text{Dice}$

4.6 Training Configuration

Hyperparameter	Value	Rationale
Backbone	ResNet-50	Strong pretrained vision encoder
Feature comparison	Concatenate	More expressive than subtraction
Max epochs	80	With early stopping
Batch size	8	GPU memory constraint

Optimiser	AdamW	Best practice for modern DL
Learning rate	1e-4 → 1e-6	Cosine annealing schedule
Weight decay	1e-4	Regularisation
Early stopping	Patience = 20 epochs	On val F1; saves best checkpoint
Mixed precision	FP16 (AMP)	~2× faster, half the VRAM

5. Experiments

5.1 Quantitative Results on LEVIR-CD Test Set

The following table presents our final test-set results alongside the targets set in our project proposal:

Metric	Target	Achieved	Status
F1 Score	> 0.78	0.906	✓ Exceeded
IoU	> 0.65	0.828	✓ Exceeded
Precision	—	0.894	—
Recall	—	0.918	—

Both quantitative targets were substantially exceeded. The high recall (0.918) indicates the model finds nearly all changed pixels; the high precision (0.894) confirms it avoids flooding the output with false positives on stable areas.

5.2 Critical Fix: The Metrics Bug

An important engineering insight emerged during development. An early version of the evaluation loop computed F1 **per batch** and averaged the per-batch scores. This is mathematically incorrect for change detection, where approximately 25% of tiles contain **zero change pixels**. On those tiles, Precision = 0/0 = 0 by convention, and F1 = 0. Averaging these zeros with the high-F1 batches from tiles that do contain change dramatically understated true performance.

The fix — implemented in the **MetricTracker** class — accumulates raw TP, FP, and FN counts across all batches and computes a single dataset-wide F1 at the end. The impact on the same trained model, same checkpoint, same test set:

Evaluation Approach	Reported F1
Batch-averaged (incorrect)	0.664
MetricTracker — dataset-wide (correct)	0.906

The model was never actually performing poorly — the measurement was wrong. This underscores the importance of understanding your evaluation metric's assumptions before drawing conclusions from training logs.

5.3 Training Dynamics

Training and validation loss curves and validation F1 over 80 epochs are shown in Figure 1 (see supplementary materials). Key observations:

- **Fast early convergence:** Training loss drops sharply in the first 5 epochs, reflecting the power of ImageNet pretraining — the encoder arrives with strong low-level visual representations.
- **Temporary validation spike at epoch ~7:** Validation loss briefly spikes before stabilising. This likely reflects the model passing through a difficult region of the loss landscape while the decoder learns to interpret the concatenated difference features.
- **Stable convergence:** Both train and validation F1 plateau smoothly after ~epoch 20. The gap between training loss (~0.10) and validation loss (~0.19) is consistent throughout training, indicating mild but stable overfitting without catastrophic divergence.
- **Early stopping:** Training ran the full 80 epochs; validation F1 continued to improve marginally through epoch 60 before fully plateauing. The best checkpoint was saved at the epoch of peak validation F1.

5.4 Ablation: Concatenation vs. Subtraction

We compared the two feature-comparison strategies on the validation set:

Diff Mode	Val F1	Val IoU
Subtract ($F_A - F_B$)	0.881	0.803
Concatenate [$F_A \parallel F_B$]	0.906	0.828

Concatenation outperforms subtraction by ~2.5 F1 points and ~2.5 IoU points. The benefit comes from the decoder's ability to learn an asymmetric comparison — distinguishing between 'A had a building, B does not' and 'B has a building, A does not' — which pure subtraction collapses into the same signal with opposite sign.

5.5 Qualitative Results: Pakistani Sentinel-2 Imagery

Figure 2 (supplementary materials) shows the three-panel before/after/predicted-change output for sample LEVIR-CD test tiles. The model correctly isolates new building footprints while ignoring vegetation colour changes between seasons.

For the Pakistani Sentinel-2 imagery, results are more mixed. On the **Lahore urban growth** region, the model detects some areas of new construction but also produces noisy predictions across vegetated zones and irregular urban textures not well represented in LEVIR-CD. On the **2022 flood region**, flood extent detection is partially successful — large inundated open areas are flagged — but the model struggles with water/soil ambiguity in vegetated floodplain areas.

6. Conclusion

This project demonstrates that a **Siamese ResNet-50 network with an FPN decoder**, trained with combined BCE + Dice loss and correct dataset-wide metric computation, achieves strong change detection performance on the LEVIR-CD benchmark ($F1 = 0.906$, $IoU = 0.828$), substantially exceeding both stated targets. Several design choices proved critical:

- **Shared encoder weights** ensure a common feature language for fair temporal comparison.
- **Concatenation over subtraction** allows the decoder to learn richer, asymmetric comparison functions.
- **FPN skip connections** recover the spatial precision lost in downsampling.
- **Independent colour jitter** teaches robustness to temporal radiometric variation.
- **Dataset-wide MetricTracker** is essential for unbiased $F1/IoU$ reporting under class imbalance and empty tiles.

The most significant limitation is **domain transfer**: LEVIR-CD contains high-resolution (sub-metre) optical imagery of US and Chinese cities, while our Pakistani test images are Sentinel-2 at 10 m/pixel with a different urban morphology. The sensor gap — in resolution, spectral response, and geometric characteristics — degrades qualitative performance on Lahore and flood imagery. Future work should include:

- Fine-tuning on a small labelled Pakistani dataset to close the domain gap.
- Extending to semantic change detection — not just 'what changed' but 'changed from what to what' (e.g., field $\rightarrow$ building).
- Multi-temporal models using 3+ time points to detect change trajectories rather than single before/after pairs.
- Model compression (knowledge distillation) for deployment on edge devices or cloud-cost-constrained pipelines.

Overall, this work provides a clean, well-validated baseline for satellite change detection with practical relevance to monitoring Pakistani landscapes, and highlights both the power and the limits of cross-domain transfer from benchmark datasets to real-world imagery.

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